Robust Sliding Mode Control with Integral Sliding Surface of an Underactuated Rotary Hook System*

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***Abstract*— In this paper, robust Integral Sliding Mode Control (ISMC) will be proposed for an underactuated rotary hook system in order to deal with parametric uncertainties presenting in the system parameters. By using the proposed ISMC, the reaching phase encountered in the conventional SMC will be eliminated and as a consequence, the dynamic ambiguity, which occurs when the system is in the transient state forwarding to the sliding mode, can be completely avoided. Furthermore, when the system slides on the sliding surface, the constraints for controller gains will be clarified in order to ensure the robust stability of the closed-loop system. All of conclusions will be drawn through both simulation and experimental results.**

I. INTRODUCTION

At the harbor, skewing mismatch between the trucks and the containers always occur. For instance, as illustrated in the Fig. 1a, the container must be rotated a skew angle of α in clockwise direction in order to place on the truck properly. If the landing angle is not precise or oscillated, the four oval holes on the corners of container cannot be inserted to the four twist-locks mounted on the truck, therefore the transloading cycle cannot be completed. Moreover, in the case described in the Fig. 1b, the container skew angle must be corrected an amount of $180° - \alpha$ in counterclockwise direction with the purpose of making the container door located at the rare end of the truck.

In reality, boom crane uses a device so-called rotary hook to rotate the container to any desired angle. In fact, as described in Fig. 2, the spreader will grip the container by the twist-lock mechanism and also be connected with a motor shaft which is mounted inside the housing of the rotary hook. Finally, the rotary hook device will be hung below the crane boom tip by flexible wire ropes. In contrast with the over-constraint skew adjustment mechanism used in gantry crane systems, which usually only allows a maximum 10 degrees change in the container skew angle, rotary hook device enables boom crane to rotate the container to an arbitrary orientation. However, along with the above advantage, after reaching the desired angle, the container will continue to oscillate rather than stop at designated skew angle due to the law between inertia force and the rope tension. In reality, the container inertia is very large and this system has very low damping ratio therefore the oscillation cannot be suppressed naturally. At present, the container orientation control through the rotary hook device is manually handled by highly experienced and skilled operator because of the under-actuated characteristic of discussed system and as a result, the transfer process is not particularly time efficient. Therefore, in order to reduce the workload for crane operator as well as significantly decrease the transloading duty cycle time, an automatic controller should be developed for the rotary hook system to perform two tasks simultaneously: driving the payload to the desired skew angle and suppressing the residual skew oscillation caused by flexible ropes. Despite the fact that many anti-swing controllers have been proposed for boom crane [1]–[8], or overhead crane system [9]–[18], there are only a few studies paying attention to the payload anti-skew oscillation controllers [19]. The very first work on modeling and designing automatic controllers for the rotary hook system should be credited to [20]. In [20], the mathematical model of the rotary hook system is derived

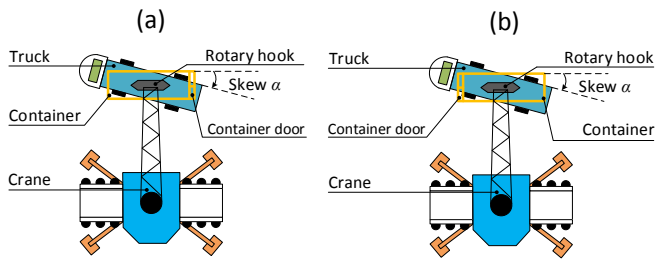

Fig. 1. Necessity of the payload orientation control

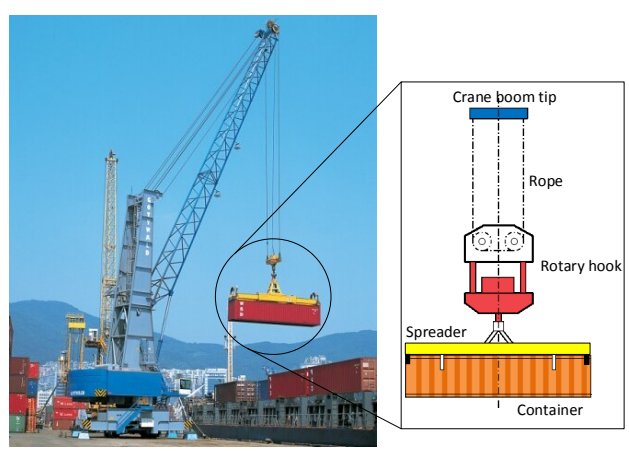

Fig. 2. Boom crane equipped with the rotary hook device

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along with the development of two controller types: time-optimum and state feedback. Both controllers are designed with the assumption that the system natural frequency is perfectly known, which is unrealistic in practice. Full-scaled experiments are only conducted to verify the effectiveness of the state feedback controller because the time-optimum control is extremely sensitive to the parameters imprecision. In [21], Schaper et al. proposed a 2-DOF skew control with novel trajectory generation to generate the nominal control law and a state feedback controller is utilized to ensure any deviation of the system outputs to the reference trajectory decay asymptotically. The performance of control system is validated with test drives on a full-size mobile harbor boom crane. However, the robustness of aforementioned controllers are not clearly verified in both simulation and experiment.

In reality, the main difficulty in designing an appropriate controller for the rotary hook system lying on the fact that the system parameters, including payload mass, payload inertia and rope length, vary in wide regions. For instance, in the full-scaled rotary hook system, the payload mass, payload inertia (in the container case) and rope length vary between 10 - 50 tons, 38103 - 819512 kgm^2, and 10 - 50 m respectively. Hence, without the robustness consideration, the controllers will even cause unstable to the closed-loop system under such large parametric uncertainties. Recently, an adaptive input shaping control [22] is developed in order to deal with the parametric uncertainties issue in the rotary hook system. The algebraic estimator is the central of the scheme which allows the conventional Zero Vibration (ZV) input shaper robust to any variation range of natural frequency. Effectiveness of the adaptive input shaping algorithm is verified through simulation results, which give a superior performance in comparison with the built-in robust Specified Insensitivity (SI) input shaper. However, this kind of control system cannot tolerate any external disturbance because adaptive input shaping is indeed an open-loop controller. From our best of knowledge, none of existing researches consider a robust feedback controller for the rotary hook system. Therefore, in this paper, we will propose a robust sliding mode control with an integral sliding surface in order to drive the payload to the desired skew angle whilst simultaneously suppress the residual skew oscillation. Moreover, the parametric uncertainties of system parameters will be elaborately considered in the controller design process in order to ensure the robust stability of closed-loop system.

In [23], a higher order sliding mode control based on integral sliding mode is discussed. The control scheme is introduced with two parts: the first part is a nominal control law to stabilize the input-output dynamic without uncertainty and the second part is to reject bounded uncertainties. An analogous approach with [23] is reported in [24], where a synthesized ISMC is proposed. Recently, disturbance observer based ISMC is theoretically developed and practically implemented in [25], [26]. Zhang et al. [27] proposed an excellent robust linear ISMC for an offshore steel jacket platform. Based on the direct Lyapunov method, the LMI constraint on controller gains is drawn to sufficiently ensure the closed-loop robust stability. However, the sliding mode does not exist at the beginning of time hence it does not make sense in the usage of the integral sliding surface. In the similar manner to [27], this paper will derive the bounds for ISMC gains in order to guarantee the robust stability of uncertain system in the presence of parametric uncertainties. However, in opposition to [27], an indirect Lyapunov method will be employed to obtain algebraic inequality constraints for the controller gains, which enable us to avoid the LMI optimization encountered in [27]. It should be noted that LMI optimization sometimes would fail to produce feasible solution.

The paper is organized as follows. Firstly, modeling of the rotary hook system is introduced and afterwards validated on the experimental apparatus. Next, a robust sliding mode controller with an integral sliding surface will be proposed, which together with appropriate controller gain constraints, will enable us to achieve two objectives simultaneously: (i) forcing the system slide on the sliding surface at the very beginning of time, (ii) forcing all controlled variables converge to their reference values regardless the parametric uncertainties in all system parameters. In order to demonstrate the effectiveness and robustness of proposed ISMC, various experiments will be conducted on the experimental testbed. It is later observed that the experimental results are in total agreement with the simulation results thus the gain tuning activities can be avoided completely when the proposed ISMC is applied to the real-size rotary hook in use at the harbor. Finally, some conclusion remarks and future works are drawn.

II. Modeling validation

The mathematical modeling of the rotary hook system is described in Fig. 3, where the skew motion of the rotary hook is resulted from the rotation of the payload. Table I provides the definitions of all system parameters.

By neglecting the hoisting motion, based on the Euler-Lagrange method, nonlinear model of the rotary hook system used in this paper is given as

$$\ddot{\theta} = f + bu, \tag{1}$$

where

$$f = \frac{1}{I_L + I_H + \dfrac{mR^4 \sin^2\theta}{a^2}} \times \left[\frac{-mR^2 \sin\theta}{a}\left(\frac{R^4\dot{\theta}^2 \sin^2\theta}{a^3} + \frac{R^2\dot{\theta}^2\cos\theta}{a} + g\right) - c\dot{\theta}\right]$$

$$b = \frac{I_L}{I_L + I_H + \dfrac{mR^4 \sin^2\theta}{a^2}}$$

$$u = \ddot{\varphi},$$

and $a = \left(l^2 - 2R^2\left(1 - \cos\theta\right)\right)^{1/2}$.

The system has only one control input (i.e. the acceleration generated by a servo motor) and two control outputs (i.e. $\varphi \to \varphi_d$, $\theta \to 0$ where φ_d is the reference angle of the

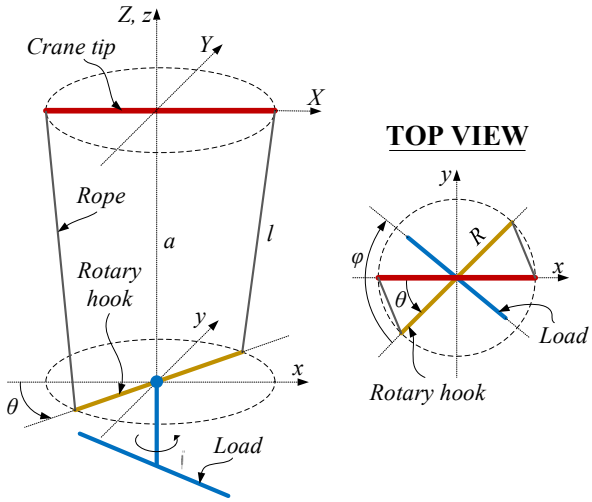

Fig. 3. Rotary hook system modeling.

TABLE I

PARAMETERS OF THE MINIATURE ROTARY HOOK SYSTEM

Symbol	Description	Range
$\varphi(t)$	Relative angle of the load and the hook (rad)	–
$\theta(t)$	Absolute angle of the hook versus x axis (rad)	–
I_L	Inertia of the payload (kgm^2)	[2.64, 41.89]
I_H	Inertia of the hook (kgm^2)	[1, 1.5]
m	Total mass of the load and the hook (kg)	[77.39, 207]
R	Skewing radius (m)	0.5
l	Rope length (m)	[2.5, 5.5]
c	Friction coefficient (Ns/rad)	[0.1, 0.5]
g	Gravity acceleration (m/s^2)	9.81

payload), therefore clearly, rotary hook is an under-actuated system.

With the purpose of experimentally verifying the robustness of ISMC, the experiment apparatus is designed such that the payload mass m, payload inertia I_L can be adjusted by adding/reducing the number of payload disks installed on the hanger (Fig. 4). Furthermore, the rope length l can

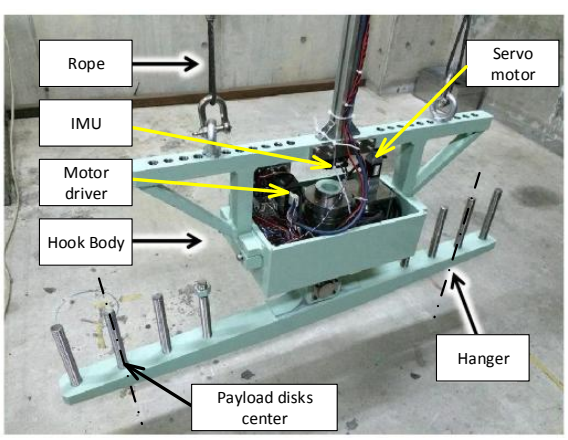

Fig. 4. Experimental testbed structure.

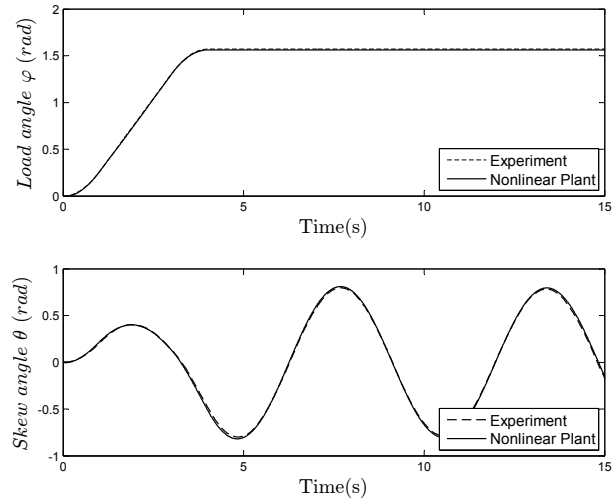

Fig. 5. Model validation in the nominal case.

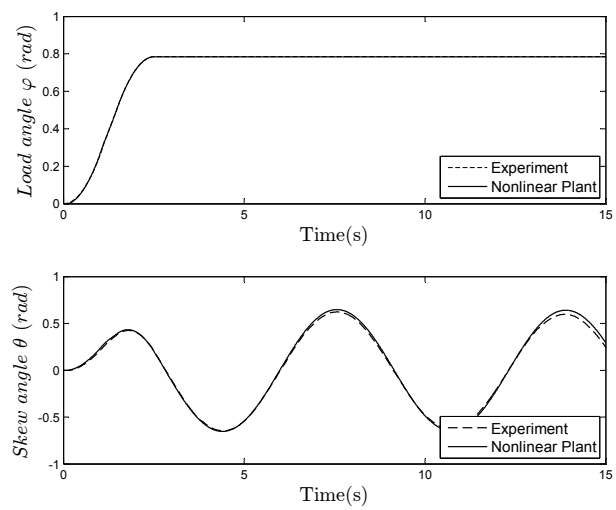

Fig. 6. Model validation in the maximum case.

also be varied by changing to another rope in different length manually. The actual variation range of all system parameters are given in Table I. Note that the nominal system parameters will be chosen at the average of their uncertain ranges, therefore the variation percentages of m, I_L, I_H, l, and c are $\pm 45.6\%$, $\pm 88.1\%$, $\pm 20\%$, $\pm 37.5\%$, and $\pm 66.67\%$ respectively. In real working process, all of aforementioned system parameters cannot be measured hence the ISMC should be designed to have an adequate robustness under such heavy parametric uncertainties condition.

In this paper, it should be clarified that "maximum case" refers to the case that the maximum number of payload disks are installed on the hanger thus m and I_L achieve their maximum values, similarly "minimum case" refers to the case that no payload disk is installed hence m and I_L achieve their minimum values (in above discussed cases, all other parameters remain at their nominal values), and finally "nominal case" corresponds to the case that all of system parameters are at their nominal values.

The nonlinear model of the rotary hook system will be

validated in two cases: nominal and maximum case. In each case, the actual system parameters of the experimental testbed will be supplied for the nonlinear model given by (1) in the simulation. The results are shown in Fig. 5 and Fig. 6 respectively. It is observed that there are strong agreements between the experimental data and the mathematical model in both cases. Therefore it is concluded that the nonlinear model (1) can describe accurately the dynamic behavior of the rotary hook system.

III. Integral Sliding Mode Control Design

A. Sliding surface and sliding mode control law design

Denote the tracking error as $e = \varphi - \varphi_d$, where φ_d is the reference angle of the payload, let us define the integral sliding function as follows

$$s(t) = k_1\dot{e}(t) + k_2 e(t) + k_3\int_0^t e(\tau)d\tau + k_4\theta(t) + k_5, \quad (2)$$

where $k_1, \ldots, k_5$ are real numbers and they will be constrained in the posterior design process. By adding the constant term k_5 into the sliding function, it is expected to completely eliminate the reaching phase (i.e. forcing the system slide on the sliding surface at the very beginning of time), therefore the constraint $s(t = 0) = 0$ should be satisfied which implies that

$$k_5 = -k_1\dot{e}(0) - k_2 e(0) - k_4\theta(0). \quad (3)$$

By assigning $\dot{s} = -Ksign(s)$ where K is a strictly positive real number and assuming the reference trajectory order is less than two (i.e. $\ddot{\varphi}_d = \dddot{\varphi}_d = \ldots = 0$), the sliding mode control law can be derived as follows

$$u = -k_1^{-1}\left(k_2\dot{e} + k_3 e + k_4\dot{\theta} + Ksign(s)\right). \quad (4)$$

In order to verify the ability of the sliding mode control law (4) in maintaining the system to slide on the sliding surface $s = 0$, the conventional Lyapunov candidate $V = 0.5s^2$ is employed, therefore

$$\dot{V} = \dot{s}s = \left(k_1 u + k_2\dot{e} + k_3 e + k_4\dot{\theta}\right)s. \quad (5)$$

Substituting the control law (4) into (5), we will obtain $\dot{V} = -K|s| \leq -\eta|s|$, where η is a sufficient small positive real number which satisfies $\eta \leq K$, hence the sliding condition is guaranteed [28], and as a consequence, the sliding surface $s = 0$ becomes an invariant set. Note that, with the choice of k_5 as in (3) and the usage of sliding mode control law (4), the system will be attracted to the invariant set $s = 0$ at $t = 0$ hence the reaching phase is now actually eliminated.

B. Robust stability analysis

In contrast with the fully-actuated case, because the current sliding surface s is a combination of the tracking error e and the residual hook skew angle θ terms thus it is not automatically ensured that on the sliding manifold $s = 0$, all of controlled variables will converge to their references (i.e. $\varphi \to \varphi_d$, $\dot{\varphi} \to \dot{\varphi}_d$, $\theta \to 0$, $\dot{\theta} \to 0$), hence the following theorem will be proposed.

Theorem 1: For the under-actuated rotary hook system described by (1), by using the sliding mode control law (4) with an under-actuated integral siding surface (2), the sufficient condition for all control output variables converge to their references (i.e. $\varphi \to \varphi_d$, $\dot{\varphi} \to \dot{\varphi}_d$, $\theta \to 0$, $\dot{\theta} \to 0$) regardless of parametric uncertainties in all system parameters is:

$$k_1 > 0, k_2 > 0, k_3 > 0, k_4 > 0, k_3 < \frac{k_1 k_2 (\omega_n)_{\min}^2}{k_2 + k_4},$$

where $(\omega_n)_{\min}$ denotes the minimum natural frequency of uncertain rotary hook system determined based on Table I.

Remark 1: By linearizing the nonlinear model of rotary hook system in (1) around the equilibrium point, the natural frequency of the hook skew angle θ can be found as $\omega_n = \sqrt{mgR^2/\left(l(I_L + I_H)\right)}$. Furthermore, based on the parameter variation ranges given in Table I, it is easy to compute the minimum natural frequency $(\omega_n)_{\min}$ appeared in the above theorem.

Proof: When the system slides on the sliding surface $s = 0$, we also have $\dot{s} = 0$, therefore

$$\ddot{\varphi} - \ddot{\varphi}_d = -k_1^{-1}k_2(\dot{\varphi} - \dot{\varphi}_d) - k_1^{-1}k_3(\varphi - \varphi_d) - k_1^{-1}k_4\dot{\theta}. \quad (6)$$

On the sliding surface, the system remains its order and is governed by (7)

$$\begin{cases} \ddot{\theta} = f + bk_1^{-1}\left[-k_2(\dot{\varphi} - \dot{\varphi}_d) - k_3(\varphi - \varphi_d) - k_4\dot{\theta}\right] \\ \ddot{\varphi} - \ddot{\varphi}_d = -k_1^{-1}k_2(\dot{\varphi} - \dot{\varphi}_d) - k_1^{-1}k_3(\varphi - \varphi_d) - k_1^{-1}k_4\dot{\theta}. \end{cases} \quad (7)$$

By defining $x_1 = \theta$, $x_2 = \dot{\theta}$, $x_3 = \varphi - \varphi_d$, $x_4 = \dot{\varphi} - \dot{\varphi}_d$, it can be obtained that

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f - bk_1^{-1}k_4 x_2 - bk_1^{-1}k_3 x_3 - bk_1^{-1}k_2 x_4 \\ \dot{x}_3 = x_4 \\ \dot{x}_4 = -k_1^{-1}k_4 x_2 - k_1^{-1}k_3 x_3 - k_1^{-1}k_2 x_4. \end{cases} \quad (8)$$

It is easy to recognize that all controlled variables will converge to their references if the nonlinear autonomous system defined in (8) is asymptotically stable at the equilibrium point $(0, 0, 0, 0)$. The robust stability condition of the nonlinear system (8) will lead to the constraints of controller gains $k_1, \ldots, k_4$.

Based on the indirect Lyapunov method, the proof now can be further proceeded. First, define the state vector $\boldsymbol{x} = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix}^\top$ and linearized model of the nonlinear system (8) around the equilibrium point $\boldsymbol{x} = 0$ can be described as

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}, \quad (9)$$

where

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\alpha_1 & -\alpha_2 & -\alpha_3 & -\alpha_4 \\ 0 & 0 & 0 & 1 \\ 0 & -k_1^{-1}k_4 & -k_1^{-1}k_3 & -k_1^{-1}k_2 \end{bmatrix}. \quad (10)$$

The symbolized elements in the matrix $\boldsymbol{A}$ are given as

$$\alpha_1 = \omega_n^2, \alpha_2 = c_N + \bar{b}k_1^{-1}k_4, \alpha_3 = \bar{b}k_1^{-1}k_3, \alpha_4 = \bar{b}k_1^{-1}k_2,$$

where $c_N = c/(I_L + I_H)$, $\bar{b} = I_L/(I_L + I_H)$.

The characteristic equation of the linear system (9) is determined as $\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = 0$, which yields in

$$\lambda^4 + h_1\lambda^3 + h_2\lambda^2 + h_3\lambda + h_4 = 0, \tag{11}$$

therein $h_1, \ldots, h_4$ are given as

$$\begin{aligned} h_1 &= \alpha_2 + k_1^{-1}k_2 \\ h_2 &= \alpha_1 + \alpha_2 k_1^{-1}k_2 + k_1^{-1}k_3 - \alpha_4 k_1^{-1}k_4 \\ h_3 &= \alpha_1 k_1^{-1}k_2 + \alpha_2 k_1^{-1}k_3 - \alpha_3 k_1^{-1}k_4 \\ h_4 &= \alpha_1 k_1^{-1}k_3. \end{aligned}$$

According to the Hurwitz stability criterion, the sufficient condition for the uncertain system defined in (9) asymptotically stable is given by (12)

$$\begin{cases} h_i > 0 \qquad (i = 1, \ldots, 4) \\ h_1h_2 - h_3 > 0 \\ h_1h_2h_3 - h_3^2 - h_1^2h_4 > 0. \end{cases} \tag{12}$$

It is quite straightforward to show that by choosing $k_1 > 0$, $k_2 > 0$, $k_3 > 0$, $k_4 > 0$, and $k_3 < \dfrac{k_1k_2(\omega_n)^2_{\min}}{k_2 + k_4}$, all of inequalities in (12) are satisfied therefore the proof is now completed. ■

Remark 2: In order to reduce the chattering in the control signal, the $sign(s)$ function in (4) should be replaced by $sat(s)$ function which is defined in (13), therein ε is a constant related to the thickness of the boundary layer [28]

$$sat(s) = \begin{cases} 1 & \text{if} \quad |s/\varepsilon| > 1 \\ s/\varepsilon & \text{if} \quad |s/\varepsilon| < 1. \end{cases} \tag{13}$$

IV. Experimental results

In all simulation and experimental studies, the controller gains of ISMC are fixed at $k_1 = 1.5$, $k_2 = 0.8$, $k_3 = 0.17$, $k_4 = 1.5$, $K = 0.55$, and $\varepsilon = 0.05$. The performance of proposed ISMC will be verified in three cases: minimum, nominal, and maximum case where the reference payload angle φ_d is 90 degrees. The comparison results between simulations and experiments for three cases are shown in Figs. 7 - 9 respectively.

Note that the gains of ISMC used in the simulations will be applied directly to the experiment apparatus, there is no gains tuning process in our experiment. The sampling time is 25 milliseconds. It is observed that in the minimum case, the peak of the hook skew angle θ during the motion is the smallest among three cases. However, the transfer time in the minimum case is longest (11.5 seconds) in comparison with the nominal case (10.1 seconds) and maximum case (8.9 seconds). Furthermore, in the maximum case, there is the largest overshoot in the load angle φ due to the large inertia. In all three cases, the sliding function s are kept inside the neighborhood of zero bounded by the designated boundary layer with a thickness of ε.

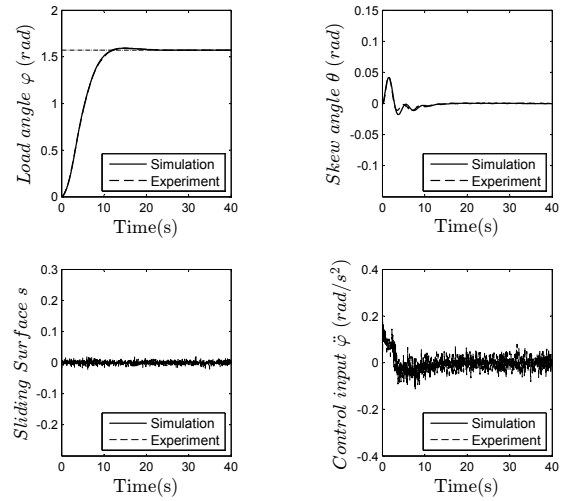

Fig. 7. Experimental result of ISMC in the minimum case.

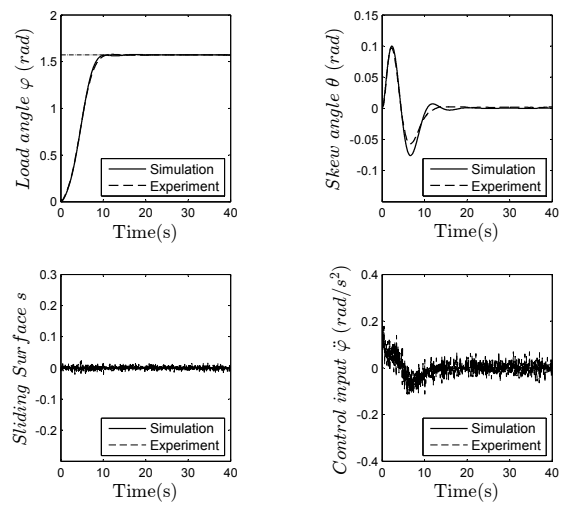

Fig. 8. Experimental result of ISMC in the nominal case.

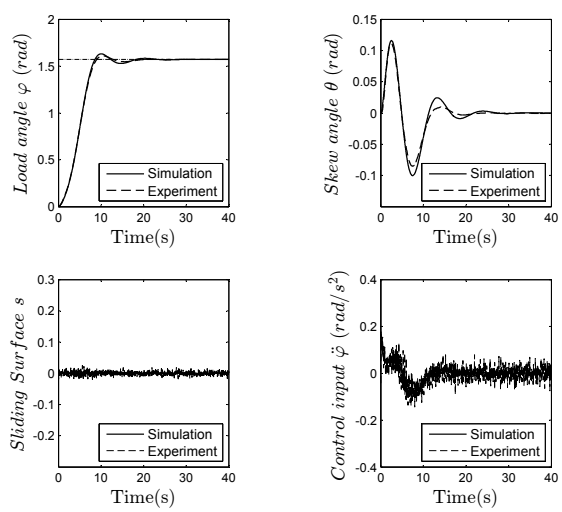

Fig. 9. Experimental result of ISMC in the maximum case.

It should be emphasized that the simulation and exper-

iment results on the experimental testbed are in complete agreement with each other, hopefully therefore, the tuning process can be avoided completely by just using the gains in simulation when the controller is implemented in real rotary hook system working at the harbor.

V. Conclusion

This paper have proposed a robust sliding mode controller with integral sliding surface for a rotary hook system. By using such discussed sliding surface type, the system can slide on the sliding surface at the very first beginning of the motion hence the dynamic ambiguity during the reaching phase can be completely avoided. All of conclusions are verified through both simulation and experimental results. Our future work is to compare the nonlinear SMC with the linear robust H_∞ control to determine the best candidate for actual implementation on the real rotary hook system in use at the harbor. Furthermore, the skew rotation dynamic of the rotary hook system will be integrated into the boom crane dynamic in order to design a comprehensive automatic controller which can suppress not only the swing angles but also residual skew oscillation of the payload simultaneously during the transfer process.

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